



Gowin GoBert

User Guide

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1 About This Guide

1.1 Purpose

This manual describes the operations of Gowin Software GoBert tool. It primarily introduces the usage of the tool and the analysis of results, aiming to help users efficiently estimate and analyze power consumption. As the software is subject to change without notice, some information may not remain relevant and may need to be adjusted according to the software that is in use.

1.2 Related Documents

The latest user guides are available on GOWINSEMI Website. You can find the related documents at www.gowinsemi.com:

- [SUG100, Gowin Software User Guide](#)
- [SUG918, Gowin Software Quick Start Guide](#)

1.3 Terminology and Abbreviations

Table 1-1 shows the abbreviations and terminology used in this manual.

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Meaning
BER	Bit Error Rate
FPGA	Field Programmable Gate Array

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly by the following ways.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 GoBert Overview

GoBert is a tool designed in-house by Gowin Semiconductor for analyzing the eye diagrams of SerDes RX signals. It helps users evaluate SerDes reception quality, thereby enhancing the performance and reliability of their designs. When using GoBert to test signal reception quality, users need to load their functional application onto the development board. Once the functional application is operating normally, the eye diagram testing can be initiated.

Functions

- Test eye diagram
- Plot HeatMap based on Bit Error Rate (BER)
- Plot Gradient
- Plot Bathtub

3 GoBert Window

GoBert window allows users to configure device parameters, line rate, and other parameters to generate the eye diagram of RX signals.

3.1 Launch GoBert Window

Click the toolbar in Gowin Software or the button " " under the "Tools" menu to launch GoBert window, as shown in Figure 3-1.

Figure 3-1 GoBert Window



3.2 Configure GoBert

3.2.1 Configure Basic Parameters

After the window is launched, Gowin Software will automatically configure serial port or parallel port downloading, as well as the JTAG Frequency, and display the current chip device information, as shown in Figure 3-2. Users can also click "Scan" to search for the "Device" again.

Figure 3-2 Configure Downloading and JTAG Frequency



3.2.2 Configure Line Rate

Users can configure the actual data reception line rate for the testing lane, as shown in Figure 3-3.

Figure 3-3 Configure Line Rate

Q0 Ln0 Line Rate	5G	Q0 Ln1 Line Rate	5G	Q0 Ln2 Line Rate	5G	Q0 Ln3 Line Rate	5G
Q1 Ln0 Line Rate	5G	Q1 Ln1 Line Rate	5G	Q1 Ln2 Line Rate	5G	Q1 Ln3 Line Rate	5G

3.2.3 Teat Lane

Users can select the testing lanes by checking checkboxes, as shown in Figure 3-4.

Figure 3-4 Select Testing Lane

☐ q0. ln0

☐ q0. ln1

☐ q0. ln2

☐ q0. ln3

☐ q1. ln0

☐ q1. ln1

☐ q1. ln2

☐ q1. ln3

3.3 Run the Testing

After completing the above configuration, click "Run Eye Monitor" to start the eye diagram test. The software will pop up a running prompt box, as shown in Figure 3-5.

Figure 3-5 Run Eye Monitor

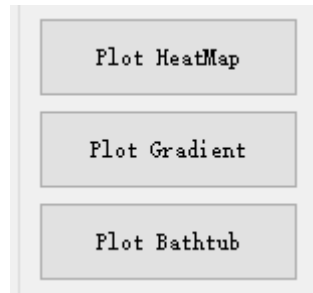
Run Eye Mon

Running...

Once this prompt box disappears, the selected lane position will display the eye diagram information.

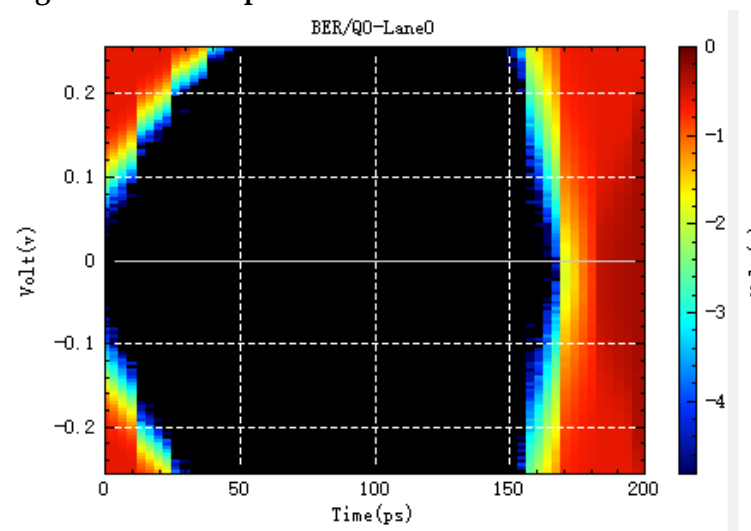
3.4 Eye Diagram Displaying

By clicking on the three options "Plot HeatMap", "Plot Gradient", and "Plot Bathtub" on the right side of the interface, different types of eye diagrams can be displayed, as shown in Figure 3-6.

Figure 3-6 Different Types of Eye Diagrams

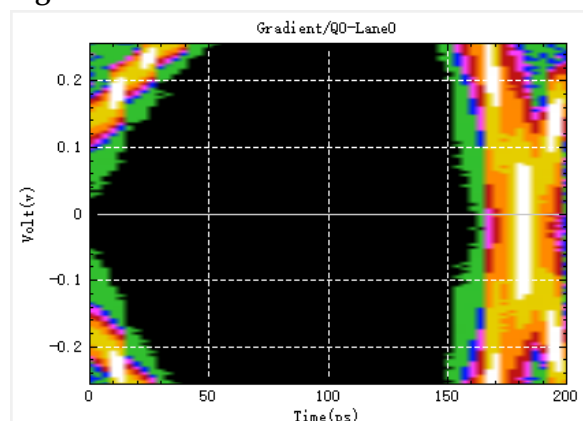
3.4.1 Plot HeatMap

The heatmap illustration is as shown in Figure 3-7. The different colors represent different bit error rates. The BER gradually decreases from red to black. Users can observe the signal quality measured in the HeatMap by referring to the relationship between the BER and the colors on the right side.

Figure 3-7 Heatmap Illustration

3.4.2 Plot Gradient

The gradient illustration is as shown in Figure 3-8, and it represents the rate of change in the grayscale values of the eye diagram, allowing for a more distinct visualization of the edges of the eye diagram.

Figure 3-8 Gradient Illustration

3.4.3 Plot Bathtub

The bathtub illustration is as shown in Figure 3-9. The vertical axis represents the BER index value, and -6 corresponds to 10^{-6} , indicating the eye width at different BER levels.

Figure 3-9 Bathtub Illustration

